

# Executive Summary

This report details the development of an automated Demand Intelligence System designed to predict future sales volumes, flag abnormal purchasing behaviors, and logically segment product inventory. By leveraging four years of historical Superstore transaction data, we built a robust forecasting engine that allows supply chain managers to confidently anticipate stock requirements. Getting ahead of the demand curve ensures the company minimizes capital tied up in dead stock while maximizing revenue by avoiding out-of-stock scenarios during the peak holiday rushes.

## Key Findings from Exploratory Data Analysis (EDA)

Our baseline analysis revealed a business model that is heavily dependent on seasonality. The Technology sector drives the highest overall revenue, while the West region provides the most stable foundation of consistent year-over-year growth. Historically, November and December generate massive revenue spikes—meaning Q4 inventory planning is the highest-leverage activity for the operations team. The average order-to-ship processing time sits at approximately 4 days consistently across all regions, indicating a healthy, uniform logistics network.

**3-Month Sales Forecast** After testing Statistical (SARIMA), Machine Learning (XGBoost), and Industry-Standard (Facebook Prophet) models, Prophet proved to be the most reliable for our data due to its ability to handle drastic holiday seasonality.

Looking at the upcoming 3-month Q4 horizon (October, November, December), the forecast projects a massive seasonal surge:

- **Month 1 (October):** A steady ramp-up in sales as early holiday shopping begins.
- **Month 2 (November):** A massive, sharp spike marking the highest sales volume of the year.
- **Month 3 (December):** Continued peak demand closing out the holiday season.
- **Confidence Ranges:** The model is highly confident in the baseline trajectory, but the confidence intervals naturally widen during November and December. This means that while we are guaranteed a massive spike, the exact ceiling of that spike is highly volatile and depends heavily on specific Black Friday promotions.

**Anomaly Detection & Root Causes** Using Isolation Forest algorithms and Z-Score deviations, the system successfully flagged abnormal sales weeks. The top 3 recurring anomalies detected are:

1. **Late November Spikes:** Consistent jumps of  $>2$  standard deviations caused by Black Friday and Cyber Monday promotions.

2. **Early January Contractions:** Unusually low sales weeks reflecting post-holiday consumer budget tightening and fatigue.
3. **Unseasonal Mid-Year Spikes:** Occasional, isolated spikes in the Office Supplies category during Q2/Q3, traced back to specific B2B corporate bulk orders rather than organic consumer demand.

**Product Demand Segmentation & Stocking Strategy** Using K-Means clustering algorithms, we segmented our catalog into 4 distinct operational buckets based on volume and volatility:

- **High Volume, Stable Demand:** Automate recurring purchase orders (POs). Maintain a tight safety stock margin to keep shelves full.
- **High Volume, High Volatility:** Aggressively scale safety stock prior to Q4, but allow inventory to run much leaner during the summer months to avoid warehouse bloat.
- **Low Volume, High Volatility:** Transition to a "Just-in-Time" (JIT) procurement model to avoid dead stock.
- **Low Volume, Low Growth:** Liquidate current inventory and phase out these underperforming sub-categories.

### **Concrete Business Recommendations**

1. **Automate Q4 Logistics Ramping:** The forecasting models confirm an unyielding upward trend every November. Supply chain teams must proactively begin scaling warehouse temporary labor and inventory procurement by late September.
2. **Regional Budget Reallocation:** Shift a larger portion of the upcoming marketing and inventory acquisition budget toward the West region's Technology sector, which our models definitively identify as the highest-yield growth area.
3. **Investigate Corporate B2B Buyers:** The anomalies found in Office Supplies indicate large, sporadic corporate purchases. Sales teams should attempt to convert these buyers into recurring contract clients to smooth out B2B demand volatility.

**System Limitations & Risks** *Risk Warning:* This forecasting system relies entirely on historical patterns. It is structurally blind to sudden external macroeconomic shocks (e.g., rapid inflation, global shipping crises, or aggressive competitor pricing drops). The business must use the interactive dashboard as an intelligence guide, but human oversight is required when external market conditions shift abruptly.